

The Theory of Twin Spaces: General Framework, Real Spacetime, and Complex Spacetime with Field Interactions

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Abstract

We present a self-contained exposition of the theory of twin spaces, from its foundations on arbitrary sets to its application to real and complex spacetimes with gauge interactions. **Part I** develops the general theory: the generation of twin operations from a base operation via gauge maps, the kernel-deformation splitting, equivariant maps between (G, H) -spaces, isomorphism theorems, and orbit-class theorems. **Part II** applies the framework to the real Minkowski spacetime $\mathbb{R}^{1,3}$, recovering Lorentz transformations, Einstein addition, and the Poincaré group as the kernel. **Part III** constructs a *complex spacetime* $\mathbb{C}^{1,3}$ equipped with the complex twin hierarchy $z_1 \odot_n z_2 = \exp^{(n)}(\log^{(n)}(z_1) + \log^{(n)}(z_2))$, where $\exp^{(n)}$ and $\log^{(n)}$ denote iterated complex exponential and logarithm. The complex hierarchy generates a countably infinite family of spacetime arithmetics parameterized by $n \in \mathbb{Z}$, each with its own isometry group. We construct a **gauge-gravity model** on $\mathbb{C}^{1,3}$ by crossing the complex twin group with gauge groups representing interactions $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))$, producing a unified framework where gravity (from the twin hierarchy) and gauge forces (from the fiber) coexist in a single (G, H) -equivariant structure.

Contents

I	General Theory of Twin Spaces	3
1	Twin Operations on an Arbitrary Set	3
1.1	The generation mechanism	3
1.2	The twin hierarchy	3
2	The Kernel-Deformation Splitting	4
3	Equivariant Maps Between Twin Spaces	5

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4	Isomorphism Theorems	6
5	Orbit-Class Theorems	6
II	Application to Real Spacetime $\mathbb{R}^{1,3}$	6
6	Minkowski Space as Twin Space	6
III	Complex Spacetime $\mathbb{C}^{1,3}$ and the Complex Twin Hierarchy	7
7	The Complex Twin Hierarchy	7
7.1	Complex logarithm and exponential	7
8	Complex Spacetime $\mathbb{C}^{1,3}$	8
9	Twin Hierarchy on Complex Spacetime	9
10	Gauge-Gravity Model on $\mathbb{C}^{1,3}$	9
10.1	The total symmetry group	9
10.2	Equivariant fields	10
10.3	The complex Einstein-Yang-Mills system	10
11	The n-Level Metric and Deformed Gravity	11
12	The Full Model: Complex Gauge-Twin-Gravity	11
13	Discussion and Open Problems	12
13.1	What is new	12
13.2	Open problems	12
14	Conclusion	13

Part I

General Theory of Twin Spaces

1 Twin Operations on an Arbitrary Set

1.1 The generation mechanism

Every construction in this paper rests on a single idea: a symmetry does not merely *preserve* an operation, it *transports* it. Conjugating a base operation α by a bijection ϕ produces a new operation $\odot_\phi$ that is abstractly isomorphic to α yet generically distinct from it as a concrete operation on the fixed set E . Letting ϕ range over a group G spreads the single operation α into an orbit of operations, and the geometry of that orbit—which ϕ fix α , and which deform it—is exactly the kernel–deformation dichotomy formalised in Section 2. This orbit is the right-coset space $\text{Ker}(\alpha) \backslash G$ that, in the companion foundational papers, carries the deformation cohomology of the structure.

Definition 1.1 (Base algebraic structure). Let (E, α) be a set E equipped with a binary operation $\alpha : E \times E \rightarrow E$.

Definition 1.2 (Gauge map). A **gauge map** is a bijection $\phi : E \rightarrow E$. The set of all gauge maps forms the symmetric group $\text{Bij}(E)$.

Definition 1.3 (Twin operation). Given (E, α) and a gauge map ϕ , the **twin operation** $\odot_\phi$ is:

$$x \odot_\phi y := \phi^{-1}(\phi(x) \alpha \phi(y)) \quad (1)$$

Equivalently: conjugate α by ϕ . The operation $\odot_\phi$ inherits all equational properties of α (associativity, commutativity, existence of identity/inverse) because conjugation preserves equations.

Theorem 1.4 (Property inheritance). *If (E, α) is a group (resp. ring, field, vector space), then $(E, \odot_\phi)$ is a group (resp. ring, field, vector space) with:*

- *Identity: $e_\phi = \phi^{-1}(e_\alpha)$ where e_α is the identity of α .*
- *Inverse: $x_\phi^{-1} = \phi^{-1}(\phi(x)_\alpha^{-1})$.*
- *Homomorphism: $\phi : (E, \odot_\phi) \rightarrow (E, \alpha)$ is an isomorphism.*

Proof. For associativity: $x \odot_\phi (y \odot_\phi z) = \phi^{-1}(\phi(x) \alpha \phi(\phi^{-1}(\phi(y) \alpha \phi(z)))) = \phi^{-1}(\phi(x) \alpha \phi(y) \alpha \phi(z)) = (x \odot_\phi y) \odot_\phi z$. The other properties follow similarly. The map $\phi : (E, \odot_\phi) \rightarrow (E, \alpha)$ satisfies $\phi(x \odot_\phi y) = \phi(x) \alpha \phi(y)$ by definition, so it is a homomorphism; being a bijection, it is an isomorphism. $\square$

1.2 The twin hierarchy

Definition 1.5 (Twin hierarchy). Given (E, α) and a fixed gauge map ϕ , the **twin hierarchy** is the family of operations $\{\odot_n\}_{n \in \mathbb{Z}}$ defined by:

$$x \odot_n y := (\phi^{-1})^{(n)}(\phi^{(n)}(x) \alpha \phi^{(n)}(y)) \quad (2)$$

where $\phi^{(n)}$ denotes the n -fold composition of ϕ (with $\phi^{(0)} = \text{id}$, $\phi^{(-n)} = (\phi^{-1})^{(n)}$). Note $\odot_0 = \alpha$ and $\odot_1 = \odot_\phi$.

Example 1.6 (The arithmetic hierarchy on $\mathbb{R}_{>0}$). Take $E = \mathbb{R}_{>0}$, $\alpha = +$ (addition), $\phi = \log$ (so that $\phi(xy) = \phi(x) + \phi(y)$):

n	$\odot_n$	Name
-1	$x \odot_{-1} y = \ln(e^x + e^y)$	LogSumExp
0	$x \odot_0 y = x + y$	Addition
1	$x \odot_1 y = x \cdot y$	Multiplication
2	$x \odot_2 y = x^{\ln y}$	Power tower

Each level generates the next via exp-conjugation. The hierarchy extends in both directions.

Example 1.7 (The Lorentz hierarchy on $(-c, c)$). Take $E = (-c, c)$, $\alpha = +$, $\phi = \operatorname{artanh}(\cdot/c)$: $v_1 \odot_1 v_2 = c \tanh(\operatorname{artanh}(v_1/c) + \operatorname{artanh}(v_2/c)) = \frac{v_1 + v_2}{1 + v_1 v_2 / c^2}$. This is Einstein velocity addition.

2 The Kernel-Deformation Splitting

Definition 2.1 (Group action on operations). Let G be a group acting on (E, α) by bijections: for each $g \in G$, $g : E \rightarrow E$ is a bijection. Each g generates a twin operation $\odot_g$ via Definition 1.3. The map $g \mapsto \odot_g$ partitions G into classes.

Definition 2.2 (Kernel). The **kernel** of the action is the stabilizer of the base operation:

$$\operatorname{Ker}(\alpha) := \{g \in G : \odot_g = \alpha\} = \{g \in G : g(x \alpha y) = g(x) \alpha g(y) \forall x, y\} \quad (3)$$

Elements of $\operatorname{Ker}(\alpha)$ are the **automorphisms** of (E, α) : they preserve the operation. In physics, these are the **symmetries**.

Definition 2.3 (Deformation space). The **deformation space** (or twin space) is the quotient:

$$\operatorname{Twin}(\alpha) := \operatorname{Ker}(\alpha) \backslash G \quad (4)$$

It parameterizes the *physically distinct* twin operations: g_1 and g_2 produce the same operation iff $g_1 g_2^{-1} \in \operatorname{Ker}(\alpha)$, i.e. iff they lie in the same *right coset* $\operatorname{Ker}(\alpha)g$. (The deformation space is thus the right-coset space $\operatorname{Ker}(\alpha) \backslash G$; it coincides with $G/\operatorname{Ker}(\alpha)$ when $\operatorname{Ker}(\alpha) \trianglelefteq G$.)

Theorem 2.4 (The splitting theorem). *The Lie algebra $\mathfrak{g}$ of G admits a splitting:*

$$\boxed{\mathfrak{g} = \underbrace{\mathfrak{k}\mathfrak{er}(\alpha)}_{\text{symmetries}} \oplus \underbrace{\mathfrak{d}\mathfrak{ef}(\alpha)}_{\text{deformations}}} \quad (5)$$

where $\mathfrak{k}\mathfrak{er}(\alpha) = \{\xi \in \mathfrak{g} : \mathcal{L}_\xi \alpha = 0\}$ (infinitesimal automorphisms) and $\mathfrak{d}\mathfrak{ef}(\alpha)$ is a complement. The kernel dimension $\kappa := \dim(\mathfrak{k}\mathfrak{er})$ counts the independent symmetries.

Proof. $\mathfrak{k}\mathfrak{er}(\alpha) = \ker(\xi \mapsto \mathcal{L}_\xi \alpha)$ is the kernel of a linear map from $\mathfrak{g}$ to the space of bilinear maps on E . It is a Lie subalgebra: the Lie derivative is a Lie-algebra homomorphism, $\mathcal{L}_{[\xi_1, \xi_2]} = [\mathcal{L}_{\xi_1}, \mathcal{L}_{\xi_2}]$, so $\mathcal{L}_{\xi_1} \alpha = \mathcal{L}_{\xi_2} \alpha = 0$ forces $\mathcal{L}_{[\xi_1, \xi_2]} \alpha = \mathcal{L}_{\xi_1} \mathcal{L}_{\xi_2} \alpha - \mathcal{L}_{\xi_2} \mathcal{L}_{\xi_1} \alpha = 0$. Any vector-space complement $\mathfrak{d}\mathfrak{ef}$ completes the splitting; note that $\mathfrak{d}\mathfrak{ef}$ is not canonical and is generally *not* a subalgebra, since the bracket of two deformations may regenerate a symmetry. $\square$

Theorem 2.5 (Kernel characterization). *The kernel $\text{Ker}(\alpha)$ is characterized by:*

- (a) $\text{Ker}(\alpha)$ is a closed subgroup of G (intersection of automorphism conditions).
- (b) $\text{Ker}(\alpha) = \bigcap_{x,y \in E} \text{Stab}(\alpha(x,y))$ (stabilizer of all products).
- (c) If E is finite-dimensional, $\kappa \leq \dim(G)$ with equality iff G acts by automorphisms.
- (d) For a metric g on a manifold: $\text{Ker}(g) = \text{Isom}(\mathcal{M}, g)$ (isometry group), and the kernel elements are exactly the Killing vector fields.

Example 2.6 (Deformations of a bilinear form are metrics). Let $E = \mathbb{R}^n$, let $\alpha = q$ be a fixed nondegenerate symmetric bilinear form, and let $G = \text{GL}(n, \mathbb{R})$ act linearly. An element g lies in the kernel iff $q(gx, gy) = q(x, y)$ for all x, y , i.e. iff $g \in O(q)$, the orthogonal group of q , of dimension $\binom{n}{2}$. The deformation space

$$\text{Twin}(q) = O(q) \backslash \text{GL}(n, \mathbb{R})$$

then has dimension $n^2 - \binom{n}{2} = \binom{n+1}{2}$, the dimension of the space $\text{Sym}^2(\mathbb{R}^n)^*$ of symmetric bilinear forms; its points are exactly the pulled-back forms $q(g \cdot, g \cdot)$ in the GL_n -orbit of q . For $n = 4$ this is 10-dimensional: the flat (pseudo-)metrics. Enlarging G from $\text{GL}(4, \mathbb{R})$ to $\text{Diff}(\mathbb{R}^{1,3})$ adjoins the translations to the kernel, producing the full Poincaré group of Theorem 6.2 and the curved metrics of general relativity in the deformation space. This is the prototype for the slogan *deformations = metrics = gravitational fields* of Part II.

3 Equivariant Maps Between Twin Spaces

Definition 3.1 ((G, H)-equivariant map). Let (E_1, α_1, G) and (E_2, α_2, H) be two spaces with group actions. A (G, H)-**equivariant map** is a pair (Φ, φ) where $\Phi : E_1 \rightarrow E_2$ is a map and $\varphi : G \rightarrow H$ is a group homomorphism, satisfying:

$$\boxed{\Phi(g \cdot x) = \varphi(g) \cdot \Phi(x) \quad \forall g \in G, x \in E_1} \tag{6}$$

The map Φ “intertwines” the actions of G and H via φ .

Theorem 3.2 (Kernel of equivariant maps). *If (Φ, φ) is (G, H)-equivariant, then:*

- (a) $\ker(\varphi) \subseteq \text{Ker}_\Phi := \{g \in G : \Phi(g \cdot x) = \Phi(x) \forall x\}$.
- (b) Φ maps G -orbits to H -orbits: $\Phi(\mathcal{O}_G(x)) \subseteq \mathcal{O}_H(\Phi(x))$.
- (c) Φ maps G -invariants to H -invariants.
- (d) If Φ is bijective and φ is an isomorphism: $(E_1, \odot_g) \cong (E_2, \odot_{\varphi(g)})$ for all g .

Proof. (a) If $g \in \ker(\varphi)$: $\Phi(g \cdot x) = \varphi(g) \cdot \Phi(x) = e_H \cdot \Phi(x) = \Phi(x)$. (b) $\Phi(g \cdot x) = \varphi(g) \cdot \Phi(x) \in \mathcal{O}_H(\Phi(x))$. (c) If $f(g \cdot x) = f(x)$ for all g , then $f \circ \Phi$ is H -invariant by the intertwining. (d) Direct from the definitions. $\square$

Definition 3.3 (Twin morphism). A **twin morphism** from $(E_1, \{\odot_g^1\}_{g \in G})$ to $(E_2, \{\odot_h^2\}_{h \in H})$ is a (G, H)-equivariant map (Φ, φ) such that for every $g \in G$:

$$\Phi(x \odot_g^1 y) = \Phi(x) \odot_{\varphi(g)}^2 \Phi(y) \tag{7}$$

The map Φ preserves the twin operations, with the level shifted by φ .

4 Isomorphism Theorems

Theorem 4.1 (First isomorphism theorem for twin spaces). *Let $(\Phi, \varphi) : (E_1, G) \rightarrow (E_2, H)$ be a surjective twin morphism. Then:*

$$E_1 / \ker(\Phi) \cong E_2 \quad \text{as twin spaces} \quad (8)$$

where $\ker(\Phi) = \{x \in E_1 : \Phi(x) = e_2\}$ and the twin operations on the quotient are: $[x] \odot_{[g]} [y] := [\Phi^{-1}(\Phi(x) \odot_{\varphi(g)}^2 \Phi(y))]$.

Proof. Standard: $\bar{\Phi} : E_1 / \ker(\Phi) \rightarrow E_2$ defined by $\bar{\Phi}([x]) = \Phi(x)$ is well-defined (since Φ is constant on cosets of $\ker(\Phi)$), bijective (surjectivity of Φ + definition of kernel), and a twin morphism (inheriting the equivariance from Φ). $\square$

Theorem 4.2 (Second isomorphism theorem). *Let $N \trianglelefteq G$ be a normal subgroup contained in $\text{Ker}(\alpha)$. Then the twin hierarchy on E descends to the quotient group G/N :*

$$(E, \{\odot_g\}_{g \in G}) \cong (E, \{\odot_{[g]}\}_{[g] \in G/N}) \quad (9)$$

In particular, twin operations indexed by elements in the same N -coset are identical.

Theorem 4.3 (Third isomorphism theorem — lattice correspondence). *Assume $\text{Ker}(\alpha) \trianglelefteq G$, so that $\text{Twin}(\alpha) = \text{Ker}(\alpha) \backslash G = G / \text{Ker}(\alpha)$ is a group. There is a bijection between:*

- (i) *Subgroups of G containing $\text{Ker}(\alpha)$, and*
- (ii) *Subgroups of $\text{Twin}(\alpha) = G / \text{Ker}(\alpha)$.*

This bijection preserves normality, indices, and quotients: if $\text{Ker}(\alpha) \subseteq K \subseteq G$, then $K / \text{Ker}(\alpha)$ is a subgroup of $\text{Twin}(\alpha)$ and $G / K \cong \text{Twin}(\alpha) / (K / \text{Ker}(\alpha))$.

5 Orbit-Class Theorems

Theorem 5.1 (Orbit-stabilizer for twin operations). *For $g \in G$ acting on (E, α) : the orbit of α under the subgroup generated by g has size: $|\mathcal{O}_{\langle g \rangle}(\alpha)| = [\langle g \rangle : \text{Stab}_{\langle g \rangle}(\alpha)]$. More generally, the number of distinct twin operations in $\{\odot_g : g \in G\}$ is $[G : \text{Ker}(\alpha)]$.*

Theorem 5.2 (Burnside class counting). *If G is finite, the number of distinct twin operations is:*

$$|\{\odot_g : g \in G\}| = [G : \text{Ker}(\alpha)] = \frac{|G|}{|\text{Ker}(\alpha)|} \quad (10)$$

Theorem 5.3 (Classification of twin structures). *Two twin operations $\odot_{g_1}$ and $\odot_{g_2}$ are equivalent under the symmetry group $\text{Ker}(\alpha)$ — i.e. related by conjugation by an automorphism of (E, α) — iff g_1 and g_2 lie in the same double coset $\text{Ker}(\alpha) \backslash G / \text{Ker}(\alpha)$. The number of such equivalence classes is the number of double cosets. (Note that every $\odot_g$ is abstractly isomorphic to α by Theorem 1.4; this $\text{Ker}(\alpha)$ -equivalence is therefore strictly finer than mere algebraic isomorphism.)*

Part II

Application to Real Spacetime $\mathbb{R}^{1,3}$

6 Minkowski Space as Twin Space

Construction 6.1 (Real spacetime). Take:

- $E = \mathbb{R}^{1,3}$ (4-dimensional real vector space).
- $\alpha = \eta$ (Minkowski metric: $\eta = \text{diag}(-c^2, 1, 1, 1)$, used as a bilinear form).
- $G = \text{Diff}(\mathbb{R}^{1,3})$ (diffeomorphism group, or $\text{GL}(4, \mathbb{R})$ for linear transformations).

Theorem 6.2 (Kernel = Poincaré). *The kernel of the Minkowski metric is the Poincaré group:*

$$\text{Ker}(\eta_c) = \text{Poinc}(c) = \text{SO}^+(1, 3) \ltimes \mathbb{R}^4 \quad (11)$$

with $\kappa = \dim(\text{Poinc}) = 10$. This is maximal: $\kappa = n(n+1)/2$ for $n = 4$.

Theorem 6.3 (Twin operations on velocities). *On the velocity space $(-c, c)$, the gauge map $\phi = \text{artanh}(v/c)$ generates the twin operation:*

$$v_1 \oplus_\phi v_2 = \frac{v_1 + v_2}{1 + v_1 v_2 / c^2} \quad (12)$$

This is Einstein velocity addition. The kernel of $\oplus_\phi$ is the identity (trivial for the abelian 1D case). The deformation space $\text{Twin}(\eta_c) = \text{Poinc}(c) \backslash \text{Diff}(\mathbb{R}^{1,3})$ (right cosets, as in Definition 2.3) parameterizes all possible metrics (= gravitational fields) on $\mathbb{R}^{1,3}$.

Remark 6.4 (GR as the deformation space). General relativity is the physics of the deformation space $\text{Twin}(\eta_c)$: a metric $g \neq \eta$ corresponds to a non-trivial twin operation $\odot_g \neq \odot_\eta$, and the Einstein equations constrain which elements of $\text{Twin}(\eta_c)$ are physically realized.

Part III

Complex Spacetime $\mathbb{C}^{1,3}$ and the Complex Twin Hierarchy

7 The Complex Twin Hierarchy

7.1 Complex logarithm and exponential

Definition 7.1 (Complex iterated logarithm and exponential). For $z \in \mathbb{C}$ with $z \neq 0$, define:

$$\log^{(0)}(z) := z, \quad \log^{(n)}(z) := \log(\log^{(n-1)}(z)) \quad (n \geq 1), \quad (13)$$

$$\exp^{(0)}(z) := z, \quad \exp^{(n)}(z) := \exp(\exp^{(n-1)}(z)) \quad (n \geq 1), \quad (14)$$

where $\log$ is the principal branch of the complex logarithm ($\text{Im}(\log z) \in (-\pi, \pi]$).

For negative n : $\log^{(-n)} := \exp^{(n)}$ and $\exp^{(-n)} := \log^{(n)}$.

Definition 7.2 (Complex twin hierarchy). The **complex twin hierarchy** on a suitable domain $\mathcal{D}_n \subseteq \mathbb{C}$ is:

$$\boxed{z_1 \odot_n z_2 := \exp^{(n)}(\log^{(n)}(z_1) + \log^{(n)}(z_2))} \quad (15)$$

where the domain $\mathcal{D}_n$ is the maximal open set on which $\log^{(n)}$ is defined (depends on n and the branch choices).

Theorem 7.3 (Properties of the complex hierarchy). *For each $n \in \mathbb{Z}$:*

- (a) $(\mathcal{D}_n, \odot_n)$ is a commutative group with identity $e_n = \exp^{(n)}(0)$ and inverse $z_n^{-1} = \exp^{(n)}(-\log^{(n)}(z))$.
- (b) The gauge map $\phi_n := \log^{(n)} : (\mathcal{D}_n, \odot_n) \rightarrow (\mathbb{C}, +)$ is an isomorphism.
- (c) Consecutive levels are related by: $z_1 \odot_{n+1} z_2 = \exp(w_1 \odot_n w_2)$ where $w_i = \log(z_i)$.
- (d) The hierarchy extends the real arithmetic hierarchy: on $\mathbb{R}_{>0}$, $\odot_0 = +$, $\odot_1 = \times$, and the higher levels give hyperoperations.

Proof. (a) $\odot_n$ is conjugate to $+$ via $\log^{(n)}$, so it inherits all group properties. (b) By definition, $\log^{(n)}(z_1 \odot_n z_2) = \log^{(n)}(z_1) + \log^{(n)}(z_2)$. (c) $z_1 \odot_{n+1} z_2 = \exp^{(n+1)}(\log^{(n+1)}(z_1) + \log^{(n+1)}(z_2)) = \exp(\exp^{(n)}(\log^{(n)}(\log(z_1)) + \log^{(n)}(\log(z_2)))) = \exp(\log(z_1) \odot_n \log(z_2))$. (d) Restriction to $\mathbb{R}_{>0}$ with $\log = \ln$: $\odot_0 = +$, $\odot_1 = \cdot$, as in Example 1.6. $\square$

Remark 7.4 (Branch cuts make $\odot_n$ a partial operation for $n \geq 1$). The group statement in Theorem 7.3(a) must be read locally. For $n \geq 1$ the principal $\log^{(n)}$ is defined only off a union of branch cuts, and the sum $\log^{(n)}(z_1) + \log^{(n)}(z_2)$ need not lie in the image of $\log^{(n)}$ on $\mathcal{D}_n$. Hence $\odot_n$ is in general only a *partial* (local) operation: closure, associativity and inverses hold on the open locus where every iterated principal logarithm involved stays within its principal domain. Globally one obtains a multivalued structure—naturally a groupoid rather than a group—while a consistent choice of branches on a simply connected domain recovers an honest commutative group, which is the sense intended in Theorem 7.3(a). The branch cuts discussed in Section 10 are precisely the boundaries of these domains.

Example 7.5 (First levels of the complex hierarchy).

n	$z_1 \odot_n z_2$	Description
0	$z_1 + z_2$	Complex addition
1	$\exp(\log z_1 + \log z_2) = z_1 \cdot z_2$	Complex multiplication
2	$\exp(\exp(\log \log z_1 + \log \log z_2))$	Complex hyperoperation
-1	$\log(\exp z_1 + \exp z_2) = \text{LogSumExp}(z_1, z_2)$	Complex soft-max

8 Complex Spacetime $\mathbb{C}^{1,3}$

Definition 8.1 (Complex Minkowski spacetime). The **complex Minkowski spacetime** is:

$$\mathbb{C}^{1,3} := \{Z = (z^0, z^1, z^2, z^3) : z^\mu \in \mathbb{C}\} \quad (16)$$

equipped with the complexified Minkowski metric:

$$\eta_{\mathbb{C}}(Z, W) := -c^2 z^0 w^0 + z^1 w^1 + z^2 w^2 + z^3 w^3 \quad (17)$$

(no complex conjugation — this is a $\mathbb{C}$ -bilinear form, not a Hermitian form). The real dimension is 8; the complex dimension is 4.

Remark 8.2 (Physical interpretation). Each coordinate $z^\mu = x^\mu + iy^\mu$ has a real part (x^μ : the usual spacetime coordinate) and an imaginary part (y^μ : a new “hidden” dimension). The real slice $\{Z : \text{Im}(Z) = 0\} \cong \mathbb{R}^{1,3}$ is ordinary Minkowski spacetime. The imaginary directions encode additional structure (internal degrees of freedom, or an extended spacetime).

Definition 8.3 (Complexified isometry group). The isometry group of $(\mathbb{C}^{1,3}, \eta_{\mathbb{C}})$ is the **complex Poincaré group**:

$$\text{Poinc}_{\mathbb{C}}(c) := \text{SO}(4, \mathbb{C}) \ltimes \mathbb{C}^4 \quad (18)$$

where $\text{SO}(4, \mathbb{C}) = \{\Lambda \in \text{GL}(4, \mathbb{C}) : \Lambda^T \eta_{\mathbb{C}} \Lambda = \eta_{\mathbb{C}}, \det \Lambda = 1\}$ is the complexified Lorentz group. The real Poincaré group embeds as the real form: $\text{Poinc}(c) \hookrightarrow \text{Poinc}_{\mathbb{C}}(c)$.

Theorem 8.4 (Complex kernel). $\text{Ker}(\eta_{\mathbb{C}}) = \text{Poinc}_{\mathbb{C}}(c)$ with $\kappa_{\mathbb{C}} = \dim_{\mathbb{C}}(\text{Poinc}_{\mathbb{C}}) = 10$ (complex dimension) = 20 (real dimension). The complexification doubles the number of symmetry generators.

9 Twin Hierarchy on Complex Spacetime

Definition 9.1 (Componentwise complex twin operations on $\mathbb{C}^{1,3}$). For $Z = (z^0, z^1, z^2, z^3)$ and $W = (w^0, w^1, w^2, w^3)$ in $\mathbb{C}^{1,3}$:

$$Z \odot_n W := (z^0 \odot_n w^0, z^1 \odot_n w^1, z^2 \odot_n w^2, z^3 \odot_n w^3) \quad (19)$$

where each component uses the complex twin operation (15). This equips $\mathbb{C}^{1,3}$ with a countably infinite family of group structures $\{(\mathbb{C}^{1,3}, \odot_n)\}_{n \in \mathbb{Z}}$.

Definition 9.2 (Complex twin group). The **complex twin group** $\mathcal{T}_{\mathbb{C}}$ is the group generated by all level transitions:

$$\mathcal{T}_{\mathbb{C}} := \langle T_n : n \in \mathbb{Z} \rangle, \quad T_n : (\mathbb{C}^{1,3}, \odot_n) \xrightarrow{\sim} (\mathbb{C}^{1,3}, \odot_{n+1}) \quad (20)$$

where $T_n(Z) = (\exp(z^0), \exp(z^1), \exp(z^2), \exp(z^3))$ is the componentwise complex exponential. $\mathcal{T}_{\mathbb{C}} \cong (\mathbb{Z}, +)$ (since $T_{n+1} \circ T_n^{-1} = T_1$ for all n).

Theorem 9.3 (Kernel at each level). At each level n , the kernel (isometry group of $\odot_n$ -compatible structures) is:

$$\text{Ker}_n := \{g \in \text{Aut}(\mathbb{C}^{1,3}) : g(Z \odot_n W) = g(Z) \odot_n g(W) \forall Z, W\} \quad (21)$$

The kernels at different levels are conjugate: $\text{Ker}_n = T_n \circ \text{Ker}_0 \circ T_n^{-1}$, where $\text{Ker}_0 = \text{Poinc}_{\mathbb{C}}(c)$ (the linear isometries at level 0). Hence all levels have the same kernel dimension ($\kappa_n = 10$ complex).

10 Gauge-Gravity Model on $\mathbb{C}^{1,3}$

10.1 The total symmetry group

Construction 10.1 (Gauge-gravity group). Define the **total symmetry group** as the semi-direct product:

$$\mathcal{G} := \underbrace{(\mathcal{T}_{\mathbb{C}} \ltimes \text{Poinc}_{\mathbb{C}}(c))}_{\text{gravity sector}} \times \underbrace{(\text{SU}(3) \times \text{SU}(2) \times \text{U}(1))}_{\text{gauge sector}} \quad (22)$$

The gravity sector acts on the *base* $\mathbb{C}^{1,3}$ (spacetime deformations); the gauge sector acts on the *fiber* (internal symmetries of fields).

Definition 10.2 (Total state space). The **total state space** is a fiber bundle:

$$\mathcal{E} = \mathbb{C}^{1,3} \times_{\mathcal{G}} F \quad (23)$$

where F is the fiber carrying the gauge representation. For the Standard Model: $F = \mathbb{C}^3 \oplus \mathbb{C}^2 \oplus \mathbb{C}$ (color triplet $\oplus$ weak doublet $\oplus$ hypercharge singlet).

10.2 Equivariant fields

Definition 10.3 ($(G_{\text{grav}}, G_{\text{gauge}})$ -equivariant field). A **physical field** on $\mathbb{C}^{1,3}$ is a section $\Psi : \mathbb{C}^{1,3} \rightarrow F$ that is equivariant under the total group:

$$\Psi(g_{\text{grav}} \cdot Z) = \rho(g_{\text{gauge}}(g_{\text{grav}})) \cdot \Psi(Z) \quad (24)$$

where $\rho : G_{\text{gauge}} \rightarrow \text{GL}(F)$ is the gauge representation and $g_{\text{gauge}}(g_{\text{grav}})$ is the gauge transformation induced by the spacetime transformation via a connection A .

Theorem 10.4 (Kernel structure of the total group). *The kernel of the total group $\mathcal{G}$ acting on $\mathcal{E}$ splits as:*

$$\text{Ker}(\mathcal{G}) = \text{Ker}_{\text{grav}} \times \text{Ker}_{\text{gauge}} \quad (25)$$

where $\text{Ker}_{\text{grav}} = \text{Poinc}_{\mathbb{C}}(c)$ (spacetime isometries) and $\text{Ker}_{\text{gauge}} \subseteq \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ (gauge transformations that leave all fields invariant). The deformation space:

$$\text{Twin}(\mathcal{G}) = \underbrace{\mathcal{T}_{\mathbb{C}}}_{\text{gravitational deformations}} \times \underbrace{(G_{\text{gauge}}/\text{Ker}_{\text{gauge}})}_{\text{gauge deformations}} \quad (26)$$

encodes all physically distinct configurations of gravity + gauge fields.

10.3 The complex Einstein-Yang-Mills system

Construction 10.5 (Field equations from the orbital pattern). Apply the three-step orbital pattern to the gauge-gravity group (22):

Step 1 (Symmetry): The total group $\mathcal{G}$ acts on $\mathcal{E} = \mathbb{C}^{1,3} \times_{\mathcal{G}} F$.

Step 2 (Conservation): The equivariance of the field equation under $\mathcal{G}$ implies:

- Gravitational conservation: $\nabla_{\mu} T^{\mu\nu} = 0$ (from $\text{Poinc}_{\mathbb{C}}$ -equivariance).
- Gauge conservation: $D_{\mu} J^{\mu a} = 0$ (from G_{gauge} -equivariance), where D_{μ} is the gauge-covariant derivative and $J^{\mu a}$ is the gauge current.

Step 3 (Uniqueness): The unique second-order, $\mathcal{G}$ -equivariant, divergence-free equations are:

$$G_{\mu\nu}[\eta_{\mathbb{C}}^{(n)}] + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu} \quad (\text{complex Einstein, at level } n), \quad (27)$$

$$D_{\mu} F^{\mu\nu a} = J^{\nu a} \quad (\text{Yang-Mills}), \quad (28)$$

where $\eta_{\mathbb{C}}^{(n)}$ is the metric induced by the $\odot_n$ operation and $F^{\mu\nu a}$ is the gauge field strength.

Remark 10.6 (What the complex hierarchy adds). The complex twin hierarchy adds structure absent from the real case:

- Level n as a new quantum number.** The twin level n parameterizes the “arithmetic” of spacetime. Transitions between levels ($n \rightarrow n+1$, implemented by the complex exponential) are additional symmetries that have no real analog.
- Branch cuts as topological defects.** The complex logarithm $\log(z)$ has a branch cut along the negative real axis. At level $n \geq 1$, the operation $\odot_n$ involves $\log^{(n)}$, which has n nested branch cuts. These branch cuts are *topological defects* in the complex spacetime — they divide $\mathbb{C}^{1,3}$ into sectors where the arithmetic changes discontinuously. They are candidates for domain walls, strings, or other topological objects.

- (iii) **The imaginary part as internal space.** Writing $z^\mu = x^\mu + iy^\mu$: the real part x^μ is ordinary spacetime; the imaginary part y^μ is a 4-dimensional “internal” space. The complex twin operations mix x^μ and y^μ , providing a geometric origin for internal symmetries.
- (iv) **Analytic continuation and CPT.** The complexified Lorentz group $\text{SO}(4, \mathbb{C})$ contains the PT transformation as an element in a different connected component. The CPT theorem (Paper 3b) is the statement that physical amplitudes can be analytically continued through the complexified group, connecting particle and antiparticle representations. The complex spacetime $\mathbb{C}^{1,3}$ makes this analytic continuation geometric.

11 The n -Level Metric and Deformed Gravity

Definition 11.1 (n -level metric). At level n , define the n -metric on $\mathbb{C}^{1,3}$ by transporting $\eta_{\mathbb{C}}$ through the gauge map $\phi_n = \log^{(n)}$:

$$\eta_{\mathbb{C}}^{(n)}(Z, W) := \eta_{\mathbb{C}}(\log^{(n)}(Z), \log^{(n)}(W)) = \sum_{\mu} \epsilon_{\mu} \log^{(n)}(z^{\mu}) \cdot \log^{(n)}(w^{\mu}) \quad (29)$$

where $\epsilon_{\mu} = (-c^2, +1, +1, +1)$.

Theorem 11.2 (Geodesics at level n). *The geodesics of $\eta_{\mathbb{C}}^{(n)}$ are the curves $Z(s)$ satisfying:*

$$\frac{d^2}{ds^2} \log^{(n)}(z^{\mu}(s)) = 0 \quad \mu = 0, 1, 2, 3 \quad (30)$$

i.e., $\log^{(n)}(z^{\mu}(s)) = A^{\mu}s + B^{\mu}$ (straight lines in the $\log^{(n)}$ -coordinate). In the original coordinate:

$$z^{\mu}(s) = \exp^{(n)}(A^{\mu}s + B^{\mu}) \quad (31)$$

At level 0: straight lines (SR). At level 1: $z^{\mu}(s) = e^{A^{\mu}s + B^{\mu}}$ (exponential curves). At level 2: double-exponential curves. Each level has qualitatively different geodesic structure.

Theorem 11.3 (Force hierarchy from level nesting). *The transition from level n to level $n+1$ introduces a “deformation force” — the deviation of $\odot_{n+1}$ -geodesics from $\odot_n$ -geodesics. The strength of this force scales as:*

$$F_{n \rightarrow n+1} \sim \frac{d}{ds} [\exp^{(n+1)}(As + B) - \exp^{(n)}(As + B)] \quad (32)$$

For large n : $\exp^{(n)}$ grows faster than $\exp^{(n-1)}$, so higher levels produce stronger deformation forces. This suggests a hierarchy of forces indexed by the twin level n , with gravity ($n = 0 \rightarrow 1$) as the weakest and higher levels corresponding to stronger interactions.

12 The Full Model: Complex Gauge-Twin-Gravity

Construction 12.1 (The unified model). The complete model on $\mathbb{C}^{1,3}$ has:

Base manifold: $\mathbb{C}^{1,3}$ with the complex twin hierarchy $\{\odot_n\}_{n \in \mathbb{Z}}$.

Symmetry group:

$$\mathcal{G}_{\text{full}} = \underbrace{\mathbb{Z}}_{\text{level}} \times \underbrace{\text{Poinc}_{\mathbb{C}}(c)}_{\text{spacetime}} \times \underbrace{\text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y}_{\text{gauge}} \quad (33)$$

Fiber: $F = \bigoplus_{\text{particles}} V_{(m,s)}^{(\mathbf{r}_c, \mathbf{r}_w, Y)}$ where $V_{(m,s)}$ is the Poincaré representation (mass, spin) and $(\mathbf{r}_c, \mathbf{r}_w, Y)$ are the gauge quantum numbers (color, weak isospin, hypercharge).

Fields: Sections $\Psi : \mathbb{C}^{1,3} \rightarrow F$ equivariant under $\mathcal{G}_{\text{full}}$.

Equations:

- (i) **Gravity:** Complex Einstein equations (27) on $(\mathbb{C}^{1,3}, \eta_{\mathbb{C}}^{(n)})$ at level n .
- (ii) **Gauge:** Yang-Mills equations (28) for the connection on F .
- (iii) **Matter:** Dirac/Klein-Gordon equations on $\mathbb{C}^{1,3}$ with gauge coupling.
- (iv) **Level coupling:** Transition operators T_n mixing different levels.

Real-slice recovery: Restricting to $\text{Im}(Z) = 0$ and level $n = 0$:

- $\mathbb{C}^{1,3}|_{\text{Re}} = \mathbb{R}^{1,3}$ (Minkowski).
- $\text{Poinc}_{\mathbb{C}}|_{\text{Re}} = \text{Poinc}(c)$.
- $\eta_{\mathbb{C}}^{(0)}|_{\text{Re}} = \eta$ (standard metric).
- The model reduces to the Standard Model on flat spacetime.

Theorem 12.2 (Consistency of the model). *The full model satisfies:*

- (a) **Real-slice consistency:** On $\mathbb{R}^{1,3} \subset \mathbb{C}^{1,3}$ at level 0, all equations reduce to the standard equations of physics (SR, GR, QM, QFT, SM).
- (b) **Level-independence of the kernel:** $\text{Ker}_n \cong \text{Ker}_0 = \text{Poinc}_{\mathbb{C}}(c)$ for all n (Theorem 9.3).
- (c) **Equivariance:** The field equations are $\mathcal{G}_{\text{full}}$ -equivariant by construction.
- (d) **Conservation:** The $\text{Poinc}_{\mathbb{C}}$ -equivariance implies $\nabla_{\mu} T^{\mu\nu} = 0$; the gauge equivariance implies $D_{\mu} J^{\mu a} = 0$.

13 Discussion and Open Problems

13.1 What is new

- (i) The complex twin hierarchy (15) generates a countably infinite family of spacetime arithmetics on $\mathbb{C}^{1,3}$. Each level n gives a different geodesic structure, a different metric, and a different notion of “straight line.”
- (ii) The branch cuts of $\log^{(n)}$ introduce topological defects in the complex spacetime that have no analog in the real case. These are candidates for physical objects (domain walls, cosmic strings).
- (iii) The level n provides a new discrete quantum number that indexes the “arithmetic” of spacetime. The force hierarchy (Theorem 11.3) suggests that different fundamental forces may correspond to different twin levels.
- (iv) The gauge-gravity crossing (22) provides a geometric framework where gravity (from the twin hierarchy on the base) and gauge forces (from the fiber) coexist in a single equivariant structure.

13.2 Open problems

- (i) **Branch cut physics.** What is the physical interpretation of the branch cuts? Do they correspond to observable topological defects? What are the boundary conditions at the branch cuts?
- (ii) **Level quantization.** Can the twin level n be related to known quantum numbers (baryon number, lepton number, generation)? The discrete nature of $n \in \mathbb{Z}$ is suggestive.

- (iii) **Imaginary dimensions.** What is the physical role of the imaginary parts $y^\mu = \text{Im}(z^\mu)$? Are they compactified (as in Kaluza-Klein)? Are they unobservable (as in complexified GR)?
- (iv) **Unitarity on $\mathbb{C}^{1,3}$.** The $\mathbb{C}$ -bilinear metric $\eta_{\mathbb{C}}$ is not positive-definite (not even Hermitian). What is the correct notion of “probability” on complex spacetime? Is the standard Hilbert space structure adequate, or does a new inner product emerge?
- (v) **Predictions.** Does the model predict new particles, new forces, or new phenomena that are absent from the Standard Model + GR? The force hierarchy (Theorem 11.3) suggests higher-level forces, but their strength and coupling remain to be computed.

14 Conclusion

We have presented the theory of twin spaces in three layers:

Part I (General Theory): The generation mechanism $x \odot_\phi y = \phi^{-1}(\phi(x) \alpha \phi(y))$ equips any set (E, α) with a family of isomorphic algebraic structures. The kernel-deformation splitting $\mathfrak{g} = \mathfrak{ker} \oplus \mathfrak{def}$ separates symmetries from deformations. Equivariant maps, isomorphism theorems, and orbit-class theorems provide the algebraic infrastructure.

Part II (Real Spacetime): On $\mathbb{R}^{1,3}$, the kernel is the Poincaré group, the twin operations give Einstein velocity addition, and the deformation space parameterizes gravitational fields (general relativity).

Part III (Complex Spacetime): On $\mathbb{C}^{1,3}$, the complex twin hierarchy $z_1 \odot_n z_2 = \exp^{(n)}(\log^{(n)}(z_1) + \log^{(n)}(z_2))$ generates a countably infinite family of spacetime arithmetics. Crossing with the Standard Model gauge group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ produces a unified gauge-gravity model where gravity emerges from the twin hierarchy on the base and gauge forces from the fiber. The real slice at level 0 recovers all known physics.

The complex extension is speculative and exploratory. Its value lies in the questions it raises — about branch cut physics, level quantization, and the geometry of internal symmetries — rather than in definitive predictions. It illustrates the power of the twin spaces framework: the same algebraic mechanism that produces Einstein velocity addition on $\mathbb{R}$ generates an infinite hierarchy of spacetime structures on $\mathbb{C}$, with the full Standard Model gauge group as a natural companion.

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